JILLIAN MILES

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EDUCATION

CARNEGIE MELLON UNIVERSITY Pittsburgh, PA
Ph.D. in Engineering and Public Policy Exp. June-July 2026
NSF GRFP Recipient, Dunlap Award for Outstanding Qualifying Exam, Phillips and Huang Family Fellowship in Energy

THE GEORGE WASHINGTON UNIVERSITY Washington, DC
Bachelor of Science in Systems Engineering, Minors in Computer Science, Psychology May 2018
Magna Cum Laude, Clark Engineering Scholar, Ellowitz Prize Recipient, Selected Student Commencement Speaker

DOCTORAL RESEARCH EXPERIENCE

LABORATORY FOR ENERGY AND ORGANIZATIONS & WORKFORCE SUPPLY CHAIN LAB
Understanding Labor Tradeoffs between Investment Opportunities for Energy Communities July 2025-Present

- Calculating workforce demand for 8 plausible or announced investments in Southwest Pennsylvania, including data centers, manufacturing, and low carbon iron and steel production
- Evaluating tradeoffs between capital expenditure, full time jobs, and local labor force fit of these industries with energy communities to better inform local development offices of long-term workforce and salary income gains

Assessing Labor Demand of EV Battery Manufacturing Sites and U.S. Readiness April 2024-February 2026

- Determined regional labor readiness for new gigafactory-scale electric vehicle battery manufacturing sites using skill similarity metrics to evaluate the transferability of existing workforce skills into battery manufacturing roles
- Visualized regional results through mapping 400+ U.S. metro, non-metro, and county readiness and found significant rural-urban divides, with rural areas often lacking sufficient engineering talent
- Presented firm-level policy recommendations and economic development policy recommendations through 3 meetings with local workforce boards and 2 national conference audiences

Regional Workforce Impacts of Decarbonizing Iron and Steel August 2022-April 2025
Publication: https://www.pnas.org/doi/10.1073/pnas.2419294122

- Developed and used occupational similarity measures to simulate a workforce transition from a high-carbon steelmaking facility to a lower carbon process to determine characteristics of worker displacement
- Evaluated opportunities for 2000+ displaced workers outside of iron and steel in the local area and found that production workers were limited in the broader workforce based on acceptable jobs in terms of skills and wage
- Publicized results in a peer reviewed journal, PNAS, and took policy memos based on said paper to 3 state-level economic development partners and forums for consideration as well as national conference audiences

WORK EXPERIENCE

DELOITTE CONSULTING LLP – Government and Public Services Sector Arlington, VA
Strategy Consultant – Department of State May 2021-July 2022
Program: Bureau of Energy Resources – Technical Assistance on Global Mining Governance

- Supported the Government of Serbia's Ministry of Mining and Energy (MME) in the sustainable development and commercialization of the Jadar lithium-borate mine being developed by Rio Tinto
- Researched, wrote, and delivered an Economic Impact Assessment analyzing the change in various economic metrics, including GDP, industrial output, and national employment, from the Jadar mine development
- Investigated the geopolitical, funding, and infrastructure systems surrounding the mine, enabling the Serbian government to make informed policy decisions on topics like sourcing renewable energy and powering the mine
- Aided the Ugandan Ministry of Mining and the Kenyan Ministry of Mining with their decarbonization strategy, including modeling programs and projects to implement in their mining sector to reach global and regional decarbonization targets

Technology Consultant – Transportation Security Administration **December 2019-May 2021**

- Led the transition of the Recruit and Hire Candidate Cloud O&M from an incumbent contractor to Deloitte through a 3-phase transition: self-guided knowledge transfer, shadowing, and reverse-shadowing
- Managed a team of 4 developers through the resolution of defects and reported out weekly on the progress being made to learn and overtake the system, and each of the technological components
- Created the Transportation Security Equipment (TSE) Data Architecture outlining the data properties in each of 11 different types of security equipment used at airports

Data and Systems Analyst – Internal Revenue Service **July 2018-December 2019**

- Created and socialized a data strategy for the entire Enterprise Case Management Program
- Co-led the execution of the Data Strategy Legacy Analysis, a bottom-up analysis of three legacy tech systems

SKILLS

Domain Expertise: Workforce resilience, skill transferability, place-based policy, energy transition, industrial decarbonization, energy and climate policy, community-level geographic analysis
Software/Tools: Python, R, Jira, Azure DevOps/TFS**,** Simio, Vensim, Ampl, SAS, SQL, VBA, Microsoft Office Suite including Word, Excel, PowerPoint, Project, Access
Professional: Data collection & interpretation, project management, quantitative & qualitative research, technical writing

ADDITIONAL RESEARCH EXPERIENCE

GWU SYSTEMS ENGINEERING DEPT in partnership with USAID and MIT
Research Assistant, Washington, DC, and Kampala, Uganda **June 2016-May 2018**
Deliverables: https://dspace.mit.edu/handle/1721.1/127275

- Developed system mapping tools, indicators, and a system dynamics model to visualize and analyze value chains, relationships, financing options, and evidence of systemic change across USAID Feed the Future activities
- Conducted field interviews, case studies, and stakeholder trainings to collect qualitative data and demonstrate how trust, relationship-building, and financing impact agribusiness outcomes for farmers, distributors, and other actors

SELECT PUBLICATIONS AND PRESENTATIONS

- **Simulating the Workforce Impacts of Decarbonizing Integrated Steelmaking**, *First and Corresponding Author,* Proceedings of the National Academies of Sciences, May 2025
- **INFORMS Annual Meeting,** Research Presentation – "Labor Implications of the Energy Transition: Assessing Transforming and Emerging Industries" Seattle, Washington October 2024
- **Technology, Data, and Policy Summit,** Research Presentation – "Local Labor Implications of Industrial Decarbonization: A Case Study of Southwest Pennsylvania's Iron and Steel Industry", MIT July 2024
- **When the Energy Transition Comes to Town,** *First and Corresponding Author,* Published in Issues in Science and Technology, December 2023
- **Lithium: Shaping Markets for a Green Future,** *Technical Writer and Research Contributor*
  Published in Global Mining Review, March 2021

TEACHING EXPERIENCE

CMU ENGINEERING AND PUBLIC POLICY DEPARTMENT
Teaching Assistant – Introduction to Engineering and Public Policy **January 2025-May 2025**

- Designed and authored the final course project, incorporating core analytical methods to evaluate the economic, technical, and social implications of legalizing online sports betting at the state level
- Led three weekly recitation sections with ~15 students each to reinforce key concepts, guide students through assignments, and facilitate discussion

Teaching Assistant – Introduction to the Theory and Practice of Policy Analysis **August 2024-December 2024**

- Facilitated weekly discussion sections to reinforce lecture material and foster critical thinking around policy analysis frameworks